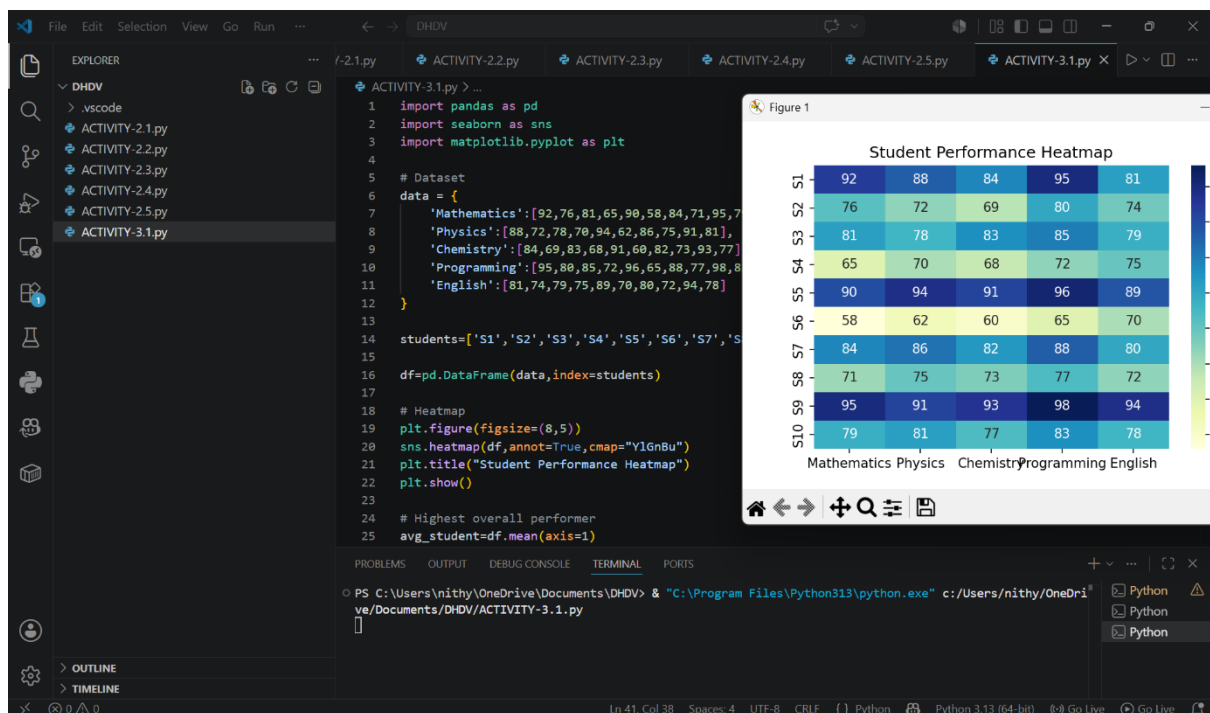


## QUESTION-1



```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
data = {
```

```
    'Mathematics': [92, 76, 81, 65, 90, 58, 84, 71, 95, 79],
```

```
    'Physics': [88, 72, 78, 70, 94, 62, 86, 75, 91, 81],
```

```
    'Chemistry': [84, 69, 83, 68, 91, 60, 82, 73, 93, 77],
```

```
    'Programming': [95, 80, 85, 72, 96, 65, 88, 77, 98, 83],
```

```
    'English': [81, 74, 79, 75, 89, 70, 80, 72, 94, 78]
```

```
}
```

```
students = ['S1', 'S2', 'S3', 'S4', 'S5', 'S6', 'S7', 'S8', 'S9', 'S10']
```

```
df = pd.DataFrame(data, index=students)
```

```
# Heatmap
```

```
plt.figure(figsize=(8, 5))
```

```
sns.heatmap(df, annot=True, cmap="YlGnBu")
```

```

plt.title("Student Performance Heatmap")

plt.show()

# Highest overall performer

avg_student=df.mean(axis=1)

print("Highest Overall Performer:",avg_student.idxmax())

# Highest average subject

avg_subject=df.mean(axis=0)

print("Highest Average Subject:",avg_subject.idxmax())

# Students needing support (<70 average)

support=avg_student[avg_student<70]

print("Students Needing Support")

print(support)

print("\nHeatmap Advantages:")

print("1. Easy comparison")

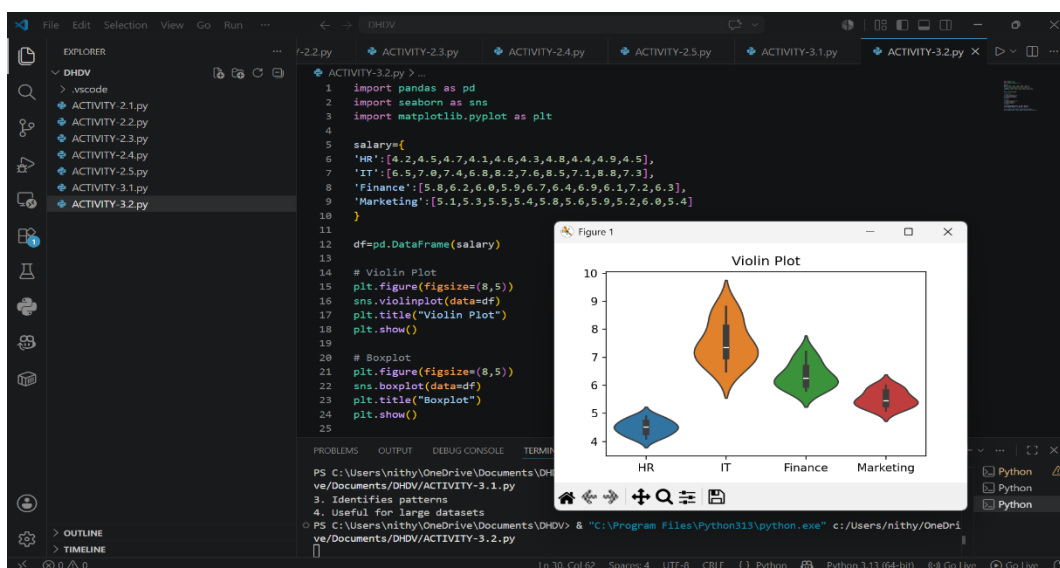
print("2. Detects high and low values quickly")

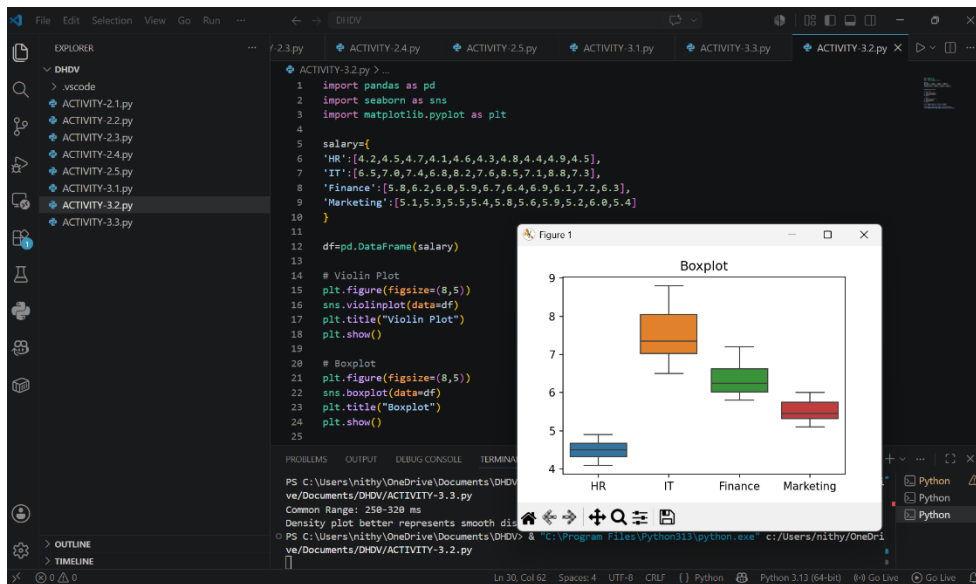
print("3. Identifies patterns")

print("4. Useful for large datasets")

```

## QUESTION-2





```
import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

salary={

'HR':[4.2,4.5,4.7,4.1,4.6,4.3,4.8,4.4,4.9,4.5],

'IT':[6.5,7.0,7.4,6.8,8.2,7.6,8.5,7.1,8.8,7.3],

'Finance':[5.8,6.2,6.0,5.9,6.7,6.4,6.9,6.1,7.2,6.3],

'Marketing':[5.1,5.3,5.5,5.4,5.8,5.6,5.9,5.2,6.0,5.4]

}

df=pd.DataFrame(salary)

# Violin Plot

plt.figure(figsize=(8,5))

sns.violinplot(data=df)

plt.title("Violin Plot")

plt.show()

# Boxplot

plt.figure(figsize=(8,5))

sns.boxplot(data=df)
```

```
plt.title("Boxplot")
```

```
plt.show()
```

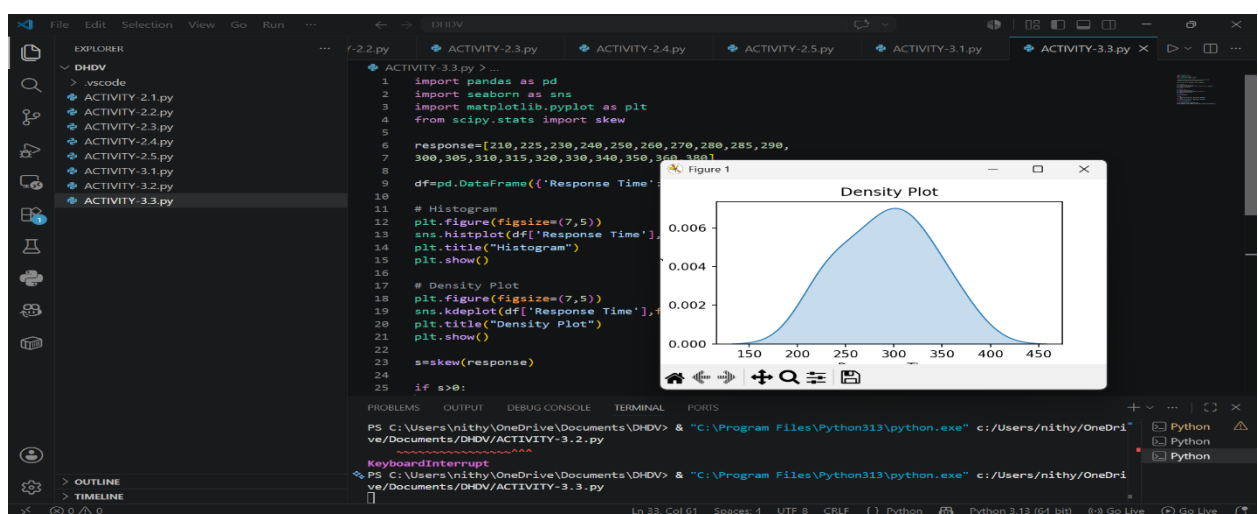
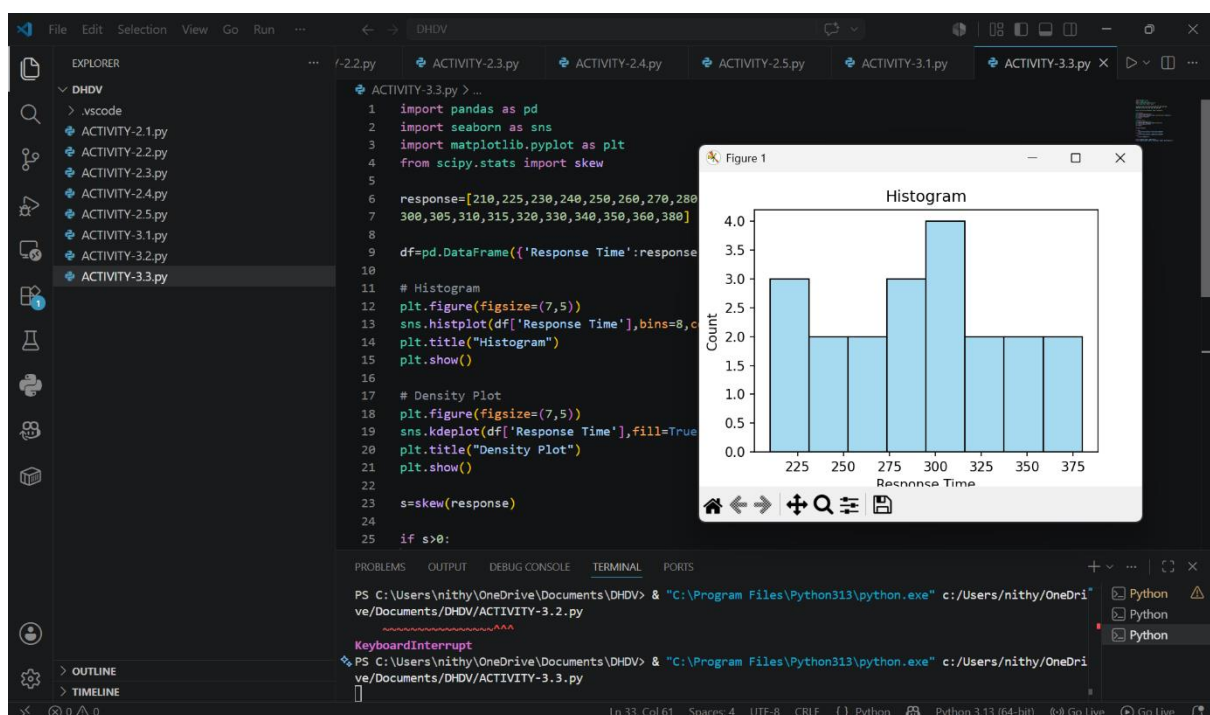
```
print("Largest Variation:",df.std().idxmax())
```

```
print("Highest Median:",df.median().idxmax())
```

```
print("\nViolin plot better shows data distribution.")
```

```
print("Boxplot better shows median, quartiles and outliers.")
```

### QUESTION-3



```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import skew

response=[210,225,230,240,250,260,270,280,285,290,
300,305,310,315,320,330,340,350,360,380]

df=pd.DataFrame({'Response Time':response})

# Histogram
plt.figure(figsize=(7,5))
sns.histplot(df['Response Time'],bins=8,color='skyblue')
plt.title("Histogram")
plt.show()

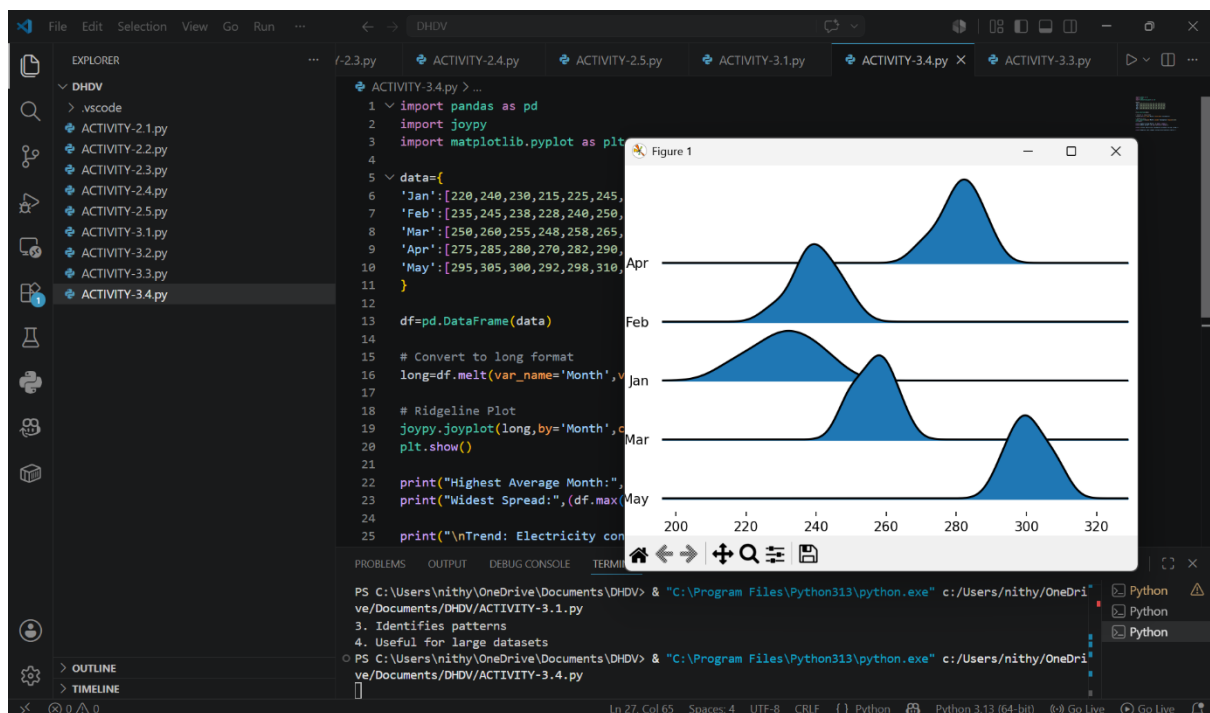
# Density Plot
plt.figure(figsize=(7,5))
sns.kdeplot(df['Response Time'],fill=True)
plt.title("Density Plot")
plt.show()

s=skew(response)

if s>0:
    print("Distribution: Positively Skewed")
elif s<0:
    print("Distribution: Negatively Skewed")
else:
    print("Symmetric")

print("Common Range: 250-320 ms")
print("Density plot better represents smooth distribution.")
```

## QUESTION-4



```
import pandas as pd
```

```
import joypy
```

```
import matplotlib.pyplot as plt
```

```
data={
```

```
'Jan': [220, 240, 230, 215, 225, 245, 235, 228, 238, 232],
```

```
'Feb': [235, 245, 238, 228, 240, 250, 242, 236, 246, 239],
```

```
'Mar': [250, 260, 255, 248, 258, 265, 259, 252, 262, 256],
```

```
'Apr': [275, 285, 280, 270, 282, 290, 284, 278, 287, 281],
```

```
'May': [295, 305, 300, 292, 298, 310, 302, 297, 307, 301]
```

```
}
```

```
df=pd.DataFrame(data)
```

```
# Convert to long format
```

```
long=df.melt(var_name='Month',value_name='Consumption')
```

```
# Ridgeline Plot
```

```
joypy.joyplot(long,by='Month',column='Consumption',figsize=(8,6))
```

```
plt.show()
```

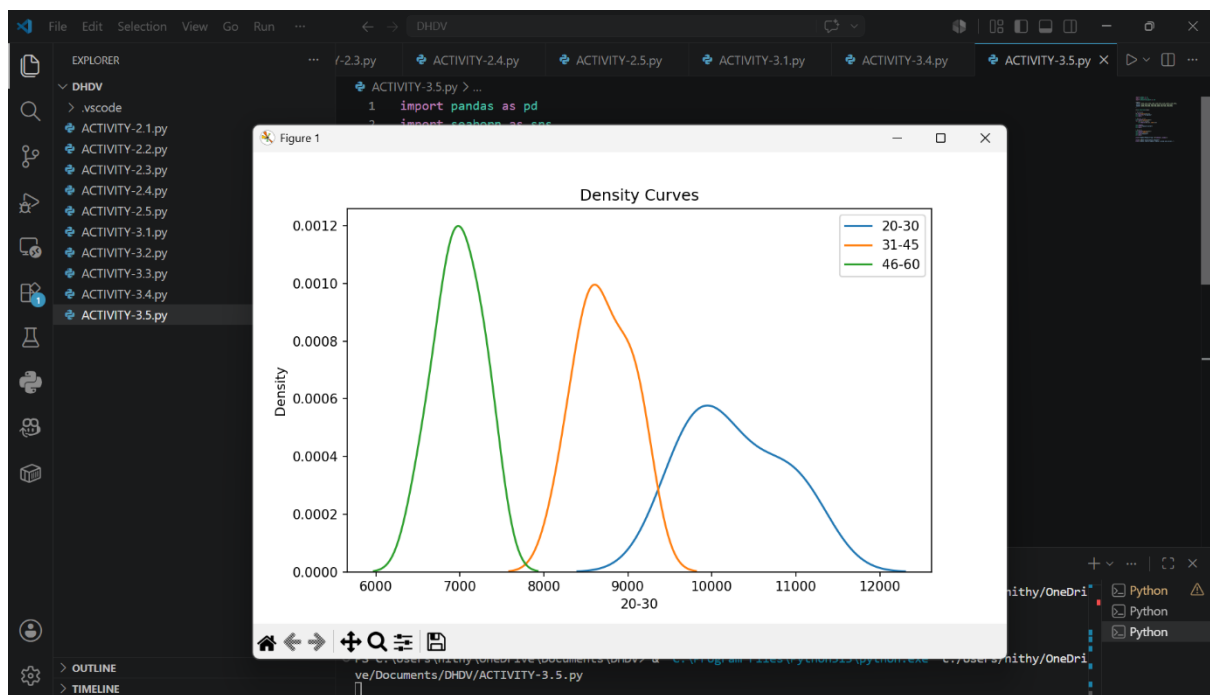
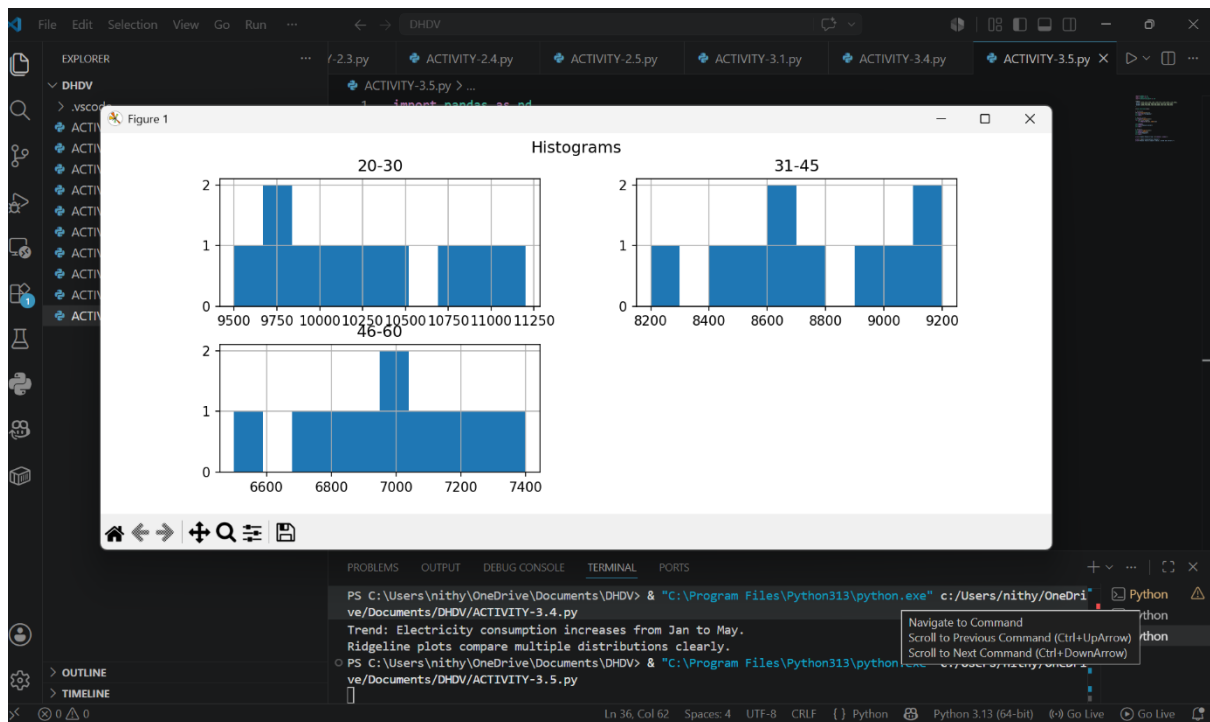
```
print("Highest Average Month:",df.mean().idxmax())

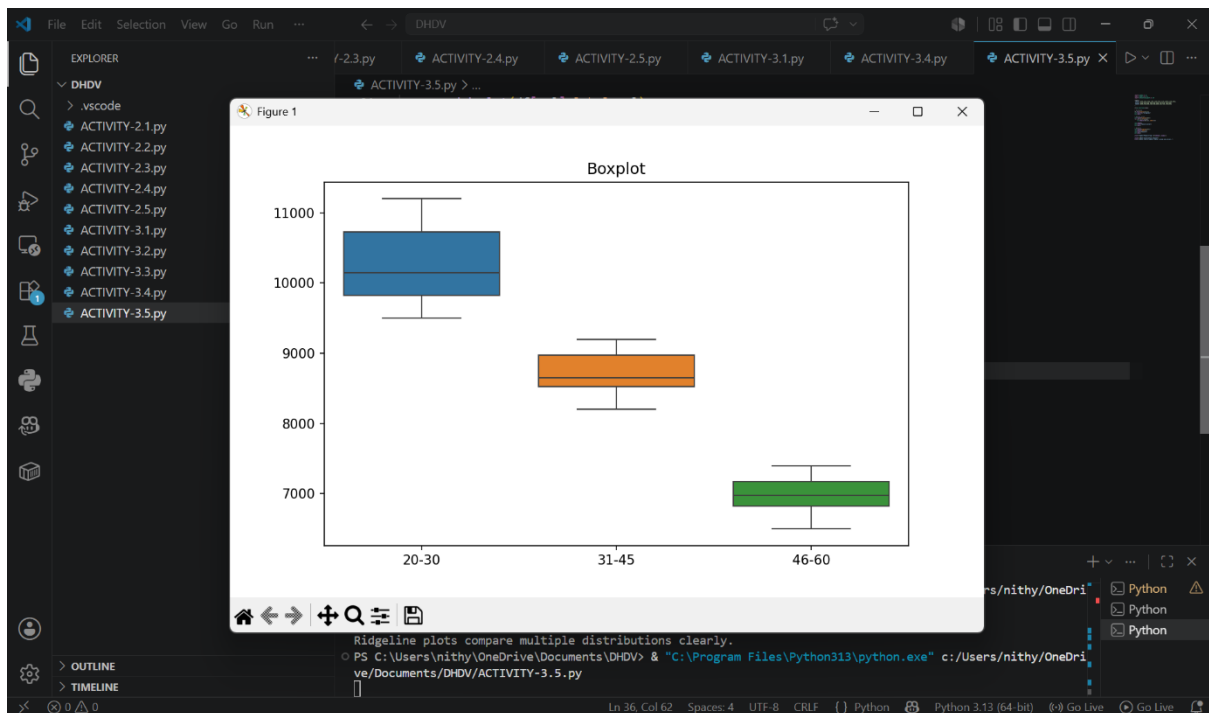
print("Widest Spread:",(df.max()-df.min()).idxmax())

print("\nTrend: Electricity consumption increases from Jan to May.")

print("Ridgeline plots compare multiple distributions clearly.")
```

## QUESTION-5





```
import pandas as pd
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
steps={
```

```
'20-30':[9500,10200,9800,11000,10500,9700,10100,10800,11200,9900],
```

```
'31-45':[8200,8600,8400,9000,8900,8500,8700,9100,9200,8600],
```

```
'46-60':[6500,6700,6900,7200,7000,6800,7100,7300,7400,6950]
```

```
}
```

```
df=pd.DataFrame(steps)
```

```
# Histograms
```

```
df.hist(figsize=(10,4))
```

```
plt.suptitle("Histograms")
```

```
plt.show()
```

```
# Density Curves
```

```
plt.figure(figsize=(8,5))
```

```
for col in df.columns:
```



```
sns.kdeplot(df[col],label=col)

plt.legend()

plt.title("Density Curves")

plt.show()

# Boxplots

plt.figure(figsize=(8,5))

sns.boxplot(data=df)

plt.title("Boxplot")

plt.show()

print("Highest Median Group:",df.median().idxmax())

print("\nBest Visualization: Boxplot")

print("Reason: Easily compares median, spread and outliers.")
```